

Aiden Van Greuning

Skills

Technical & Analytical Skills	Biomedical signal processing, data analysis, feature extraction, MATLAB, algorithm implementation, machine learning fundamentals, biomedical data interpretation, computational modeling, system characterization
Experimental, Systems & Engineering Skills	Experimental system design, embedded control systems, Arduino-based development, sensor integration, fluid mechanics and biofluid modeling, peristaltic pump systems, experimental prototyping, validation and testing, microfluidics concepts
Research, Laboratory & Professional Skills	Wet-lab techniques and cell culture, optical microscopy, biomaterials handling, protocol execution under biosafety constraints, experimental documentation, technical reporting, literature review, feasibility analysis

Education

2020–2025 **Bachelor of Engineering in Biomedical Engineering**, *Toronto Metropolitan University*

Academic Projects

Capstone: Low Cost Cerebrovascular Flow Pump (Sep. 2024–Apr. 2025) Grade: 89%	• Solely executed a capstone project typically assigned to a four-person team, designing and building a pump system (cost ~\$600, ~75% less cost compared to similar systems in professional labs) to replicate brain arterial blood flow for interventional neuroradiology training • Integrated with a 3D-printed Circle of Willis model, developing an Arduino-based control system for precise flow regulation and collecting flow/pressure data with a flow sensor and Poiseuille's Law using a glycerol–water blood analog
Signal Processing of Cerebromicrovascular Disease (Nov.–Dec. 2024) Grade: 75%	• Processed and analyzed electrocardiogram (ECG) signals from elderly diabetic patients with cerebromicrovascular disease, applying the Pan–Tompkins Algorithm for heartbeat event detection and feature extraction/analysis • Identified PQRST points and achieved 66.7% sensitivity and 62.3% validation accuracy in machine learning classification; ~20% above literature benchmarks

Experience

Apr.–Apr. 2025	Deputy Returning Officer , <i>Elections Canada</i> , Richmond Hill, ON • Verified voter ID, issued ballots and maintained accurate records for 103 electors; assisted with ballot counting, documentation and compliance reporting
Aug. 2024–Mar. 2025	Vice President of Administration (Volunteer) , <i>IEEE EMBS TMU Chapter</i> • Led and delegated tasks to a 7-member team, fostering collaboration with TMU student groups and external professionals to advance joint initiatives
Sep.–Dec. 2023	Innovation Boost Zone Apprentice , <i>PINION (Innovation Boost Zone Startup, TMU)</i> • Recruited volunteers through surveys/outreach and developed the startup's first website to support international student onboarding

Certifications

2025	Canada Good Clinical Practice (GCP)–Basic , <i>N2/CITI Program</i>
2025	Tri-Council Policy Statement (TCPS 2: CORE 2022) , <i>Government of Canada</i>